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1 PURIFYING THE AIR WE BREATHE WITH PULSED RADIO WAVES

Bengaluru-based startup Devic Earth is tackling air pollution by using pulsed radio waves to cause dust and pollen to settle down, thus cleaning the air for everyone



Pure Skies installation map

In 2008, a young cardiologist moved from the US to India in hopes of serving his country. Little did he know what lay ahead of him when he witnessed the tragic death of a 32-year-old taxi driver on the operating table during an angioplasty. The taxi driver had succumbed to a heart attack because of continuously inhaling polluted air. This incident propelled Dr Srikanth Sola to set up a team of scientists and engineers who, after ten years of research, from 2008 to 2018, came up with an answer for ambient air pollution—Devic Earth's Pure Skies.

With a background in biomedical engineering and knowledge of MRI scans, Dr Sola founded Devic Earth

in 2018 with the idea that radio wave pulses, which had earlier been used to tackle water pollution, could be used to solve the problem of air pollution.

The Bengaluru-based startup's flagship product, Pure Skies, works on pulsed Wi-Fi technology to improve air quality and can cover an area of up to a 250 to 500 metres radius outdoors or indoors, without any maintenance. The company claims to achieve a downward trend in the pollution levels within a few months across large areas.

Pure Skies has a receiver antenna attuned to the Wi-Fi spectrum (2.4 to 2.5GHz), a pulse generator, PCB boards, and a solar power sup-



Pure Skies device

ply. The device's transmitter emits rapid radio pulses that are amplified by the receiver to cover a large area. This creates an electric charge field that bipolarises microscopic pollutant particles (PM2.5 and PM10) in the air such as dust and pollen. The bipolarised particles get attracted to each other causing them to become heavier and settle down once they have picked up the charge.

"The charge is acquired by particles in this field according to their